

Vision-Based Autonomous Docking for ROVs with Relative Motion Between Dock and Vehicle *Interim Report and Project Plan*

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Abstract

Remotely Operated Vehicles (ROVs) are unoccupied, high maneuverability underwater robots operated by a human pilot offboard. They are widely used for subsea inspection, maintenance and data collection tasks across industries. However, working-class ROV operations are constrained by vessel time, weather windows and risky launch/recovery procedures, while battery powered vehicles such as AUVs and hybrid ROVs are limited by their battery endurance due to their energy capacity. To address these challenges, many recent AUVs or industry-class ROVs have explored autonomous docking to an underwater charging/data transfer station as a means to extend mission duration and reduce long-term operational costs. Most prior research, however, focuses on AUVs docking to static station fixed to a rigid structure, with relatively little work on vision-based docking to free-floating or suspended stations. This thesis investigates the feasibility of an underwater-adaptive vision system that uses a two-level strategy: active light cues for long-range guidance and passive fiducial markers for precise six degrees-of-freedom (DOF) pose estimation. This report presents the problem formulation, preliminary literature review, methodology and project plan. Preliminary work includes a basic ArUco pose-estimation under ROS 2 Humble and initial CAD modelling of a suspended docking station.

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1 Introduction

1.1 Context of Underwater Robotics and ROVs

Underwater robotic systems are widely used to offshore energy, marine science and subsea infrastructure inspection. These platforms fall into two categories: ROVs and AUVs. ROVs are tethered vehicles that remain connected to a surface vessel or platform via a long cable providing power and communications. They are typically teleoperated by a human pilot using live video and sensor feedback, and are well suited to tasks that require real-time human decision-making, complex manipulation or compliance with conservative safety procedures.

In contrast, AUVs are untethered vehicles that is powered by their own battery and rely on onboard autonomy for navigation and decision-making. They are generally used for underwater missions, such as seabed mapping, pipeline inspection or environmental monitoring. The vehicle follows a pre-planned route and returns to the surface to offload data at the end of the mission. These operations often take place in harsh environments and at considerable depth, where divers cannot safely work. The ability to extend the endurance and autonomy of such systems has therefore become an active area of research and development.

1.2 Why Docking Is Important

Docking systems address several challenges of underwater robotic operations. A subsea docking station can provide electrical power for battery recharging and a physical interface for high-bandwidth data transfer. Instead of returning to the surface vessel after each mission, a vehicle can periodically return to the dock, recharge, upload data and receive new tasking. This enables longer missions, reduces turnaround time and allows for persistent presence in the area of interest.

Docking also opens up new modes of operation. A dock can act as a local base station for multiple vehicles, support tool or payload exchange, and serve as a fixed reference point for navigation and calibration. For ROVs in particular, docking mechanisms that allow the vehicle to park safely when not in use can reduce wear on equipment and improve safety in congested subsea work sites.

1.3 Limitations of Current ROV Operations

Despite their versatility, current ROV operations face several limitations that motivate this project. Work-class ROVs typically depend on an umbilical for both power and communications. A prime example is recent REV Ocean's Aurora ROV, which weighs 4200kg and able to travel up to 6000km range [?]. However, the tether restricts operational radius, introduces hydrodynamic drag, and can snag on structures or the seabed. For smaller electric ROVs that rely on onboard batteries, endurance is directly limited by battery capacity and the power required for propulsion, lighting and sensors.

Launch and recovery require a surface vessel, crane or A-frame on the deck and a trained team. Operations are constrained by sea state and weather windows, and each deployment incurs significant mobilisation and vessel time costs. When an ROV must repeatedly return to the surface for charging or data offload, these constraints become even more pronounced.

In the research literature, much of the work on autonomous docking has focused on AUVs docking to static, seabed-mounted stations as they are the most common configuration for current AUV systems. These configurations simplify the problem by assuming a fixed dock pose relative to the environment. In contrast, ROV-related docking is often framed in terms of Tether Management Systems (TMS) or simple garages, and is generally piloted manually. Autonomous docking for ROVs to free-floating or suspended stations has received comparatively little attention.

1.4 What a Free-Floating Docking Station Is

This project considers a free-floating docking station concept rather than a fixed seabed installation. In this context, a “free-floating” dock refers to a docking structure that is suspended and mechanically connected to a surface structure via a cable or frame, but is not rigidly attached to a fixed structure. The station may be approximately stationary in a local frame, but it can move in surge, sway, heave and rotation due to waves, currents and motion due to the nature of water currents.

A free-floating station offers several advantages. It can be deployed at a convenient depth for the ROV, relocated with the vessel or between sites, and recovered for maintenance without subsea intervention. It also avoids the need for seabed foundations or long-term moorings, which can be costly and environmentally intrusive. However, the resulting relative motion between the dock and the surrounding water makes guidance and control more challenging than for a static seabed dock.

Designing a docking system for such a platform requires explicit consideration of dock dynamics, tether behaviour and the interaction between ROV and station during the final capture phase.

1.5 Why Vision-Based Docking Is Difficult

Vision-based docking relies on cameras to detect visual features on or near the docking station and derive the relative pose between the vehicle and the dock. While cameras are a common tool for visual information due to their low cost, rich information content and suitability for fine alignment, underwater imaging introduces several challenges.

Water turbidity causes light scattering and absorption, which reduce contrast, blur edges and degrades the effective visibility range. The lighting conditions underwater are highly variable: at depth there may be very low ambient light, while near the surface specular reflections/refractions and sun glint can saturate parts of the image.

Additionally, onboard computing resources on small ROVs are limited. Algorithms must operate in real time with bounded latency and power consumption. This is particularly challenging for deep learning-based approaches, which can provide strong detection performance but may require optimisation or model compression for embedded deployment.

1.6 High-Level Description of the Proposed System

To address these challenges, this thesis proposes a vision-based docking system that combines turbidity-adaptive image processing, two-level marker-based guidance and motion-compensated control. A front camera on the ROV observes a docking station equipped with both active lights and passive fiducial markers.

At longer range, the system uses LED cues as a robust, high-visibility target for coarse guidance. At close range, it transitions to passive markers such as ArUco or AprilTag high contrast patterns mounted on the docking structure to obtain its precise pose estimates via Perspective- n -Point (PnP) algorithms. Image pre-processing and enhancement are tuned to underwater conditions to maximise marker and light visibility without amplifying noise.

Pose estimates are fused over time with inertial measurements to produce a stable relative pose, which is then used by a main controller to execute a sequence of behaviours: search for the dock, approach to the dock, align with the dock axis, and final engagement into the docking structure.

1.7 Thesis Structure

This Thesis A report is organised as follows:

- Section 2 presents a literature review of existing AUV and ROV docking systems, with a particular focus on vision-based techniques, other sensing modalities and known limitations of current approaches.
- Section 3 defines the research question and hypotheses in detail, and outlines the proposed methodology, including the perception, estimation and control components.
- Section 4 describes the project-dependent preparations that had been undertaken, including preliminary ArUco-based pose estimation on real images and initial CAD modelling of the docking station concept. This section also summarises planned hardware, budget considerations and key risks.
- Finally, Section 4.3 concludes the report by summarising the current status, highlighting the main anticipated challenges, and outlining the planned work for Thesis B and Thesis C, where the full system will be implemented, tested and evaluated through tank and pool experiments using an ROV platform.

2 Literature Review

2.1 Existing Work in AUV/ROV Docking

Autonomous Underwater Vehicle (AUV) docking has been a focus of research for decades, driven by the need for long-duration missions via subsea battery recharging and data transfer. A wide range of docking guidance strategies has been explored, as summarized in recent surveys [?]. Broadly, approaches are categorised by the sensing and guidance modality. Acoustic homing using sonar beacons can guide AUVs from long range, often providing range and bearing to a dock [?]. Electromagnetic docking systems have also been tested (e.g. using low-frequency EM fields for short-range homing), notably in Feezer’s EM homing system [?]. Vision-based techniques, using cameras to detect visual markers, offer high precision in final alignment. In practice, multi-modal solutions are common: for example, an AUV might navigate toward a dock via acoustic homing signals and then switch to vision-based guidance in the final meters [?]. Dedicated docking markers can be active (such as

flashing lights or LED beacons) to increase visibility, or passive high-contrast patterns that require no power, as demonstrated in Marie’s pole design for homing detection. On the mechanical side, most docking station designs incorporate funnel-shaped guide surfaces to help capture the vehicle despite some misalignment. This combination of sensor guidance and mechanical alignment ensures a robust docking process under various conditions.

2.1.1 AUV Docking Demonstrations

Numerous projects have demonstrated AUV docking with different infrastructures and homing strategies. One of the earliest was the WHOI REMUS AUV docking system from the late 1990s, which used a Relative Acoustic Tracking system of acoustic transponders for homing into a funnel-like subsea dock (see Figure 1).

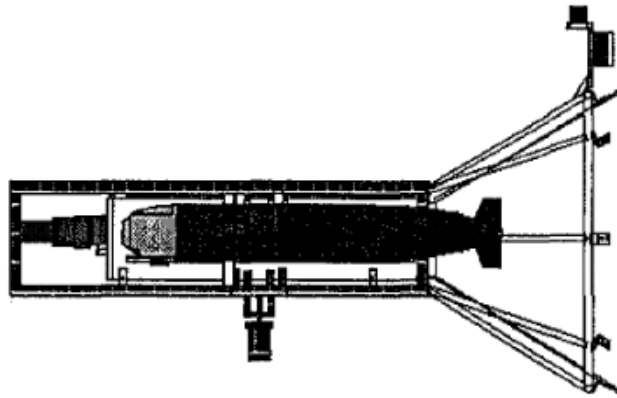


Figure 1: A side-view drawing of the WHOI REMUS AUV docking system [?].

Researchers have since tested seabed-mounted and moored docking stations to support AUV missions. For instance, Maire et al. (2009) proposed a heavy weighted docking target pole lowered to the seafloor, equipped with an acoustic beacon for long-range localisation and a visual pattern for final approach. An AUV can home to such a seafloor station, attach itself, and enter a sleep or recharge mode until the next mission window.

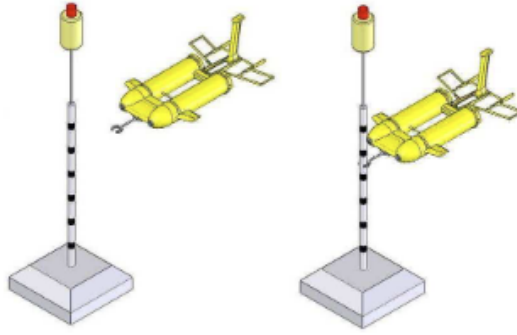


Figure 2: Marie et al.’s seabed docking station concept with acoustic and visual homing aids. (left: on approach; right: after docking)[?]

Another notable docking system design is Trslc’s moored docking station, which a cage-like docking garage was hung under a support vessel, allowing the AUV to navigate into the cage and be lifted out without surfacing on its own.

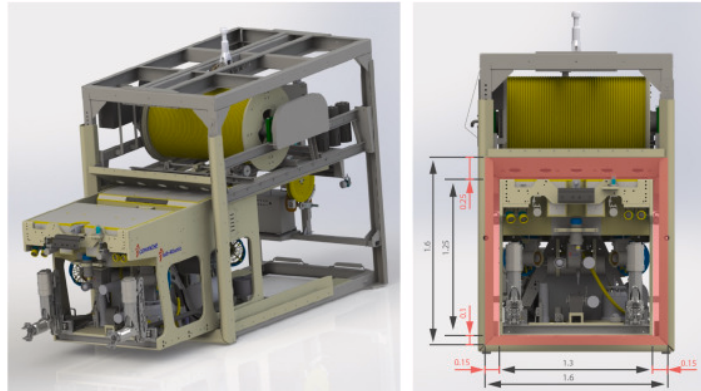


Figure 3: Trslc’s Tether Management System (TMS) docking cage for work-class ROVs.[?]

2.1.2 ROV Docking and Industrial Practice

ROVs, being tethered and human-piloted, traditionally “dock” via their TMS rather than autonomous stations. In industry, a work-class ROV is typically deployed within a TMS, essentially an underwater cage or “garage”, that is lowered from the surface. The ROV undocks to perform its task and later returns to latch into the TMS for recovery to the ship. This process is usually manual: an ROV pilot must carefully navigate the vehicle back into the moving cage, which is task made difficult by limited visibility and the unpredictability of the water current. To improve safety and extend operational envelopes, recent research has begun automating ROV docking. Notably, Trslc demonstrated the first autonomous TMS docking for a heavy work-class ROV in the North Atlantic (2019) [?]. Using a vision-based pose estimation system and the four active light markers on the TMS, their ROV was able to locate and insert itself into a standard cage without any special modifications to the hardware. This

trial, performed well in dynamic wave conditions, showed that autonomous control could consistently outperform human pilots for TMS docking, reducing the risk of collision and timing errors for such expensive equipment. The success of this approach is a major step toward “resident ROVs” that live subsea. In a different vein, researchers have also developed hybrid docking setups where an ROV acts as a mobile docking station to service AUVs. One such system (by Mitsubishi Heavy Industries) called Naminou equips an ROV with an electromagnetic coupling and wireless charger to capture an AUV in open water for battery charging [?].

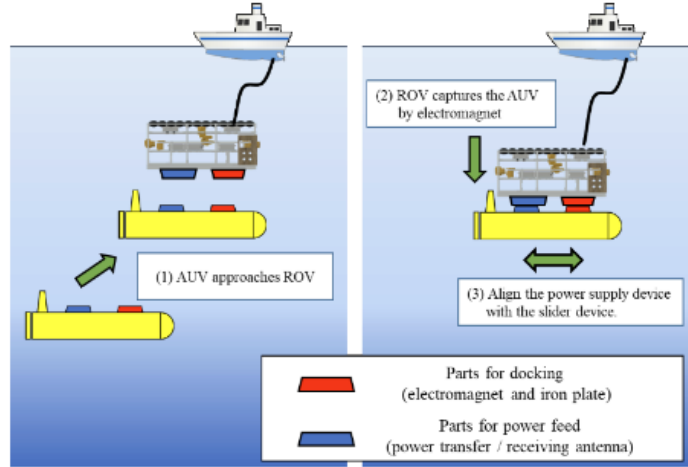


Figure 4: A flowchart of ROV-AUV docking interaction from Morinaga et al. [?]

In trials of this concept, the AUV would autonomously approach the hovering ROV using acoustic and sonar guidance, then hold position while the ROV latches onto it and aligns the power transfer coils (see Figure 4). This illustrates the breadth of ROV-involved docking solutions, from improving launch/recovery systems to enabling ROVs as docking stations for AUV support.

2.2 Vision-Based Docking Techniques

Underwater docking for AUVs relies on robust computer vision methods due to the challenges of the marine environment. Vision-based techniques provide precision in final alignment and locking phase of the docking process, complementing other sensors (like acoustics) when an AUV approaches a docking station. Key approaches include using markers, active visual beacons, deep learning for target recognition, optical flow/feature tracking for motion estimation, and specialised underwater image enhancement. This section provides an in-depth analysis of these techniques and their effectiveness under underwater conditions.

2.2.1 Existing Passive Marker-Based Docking Techniques

Markers such as AprilTags and ArUco codes are popular for robotic navigation and pose estimation. These markers are detected by extracting the four corner vertices, followed by a Perspective-n-Point

(PnP) algorithm to compute the 6-DOF pose of the marker relative to the camera. The low computation cost and geometric accuracy make them attractive for underwater docking guidance.

Passive visual markers for underwater docking are typically high-contrast black and white patterns or coded shapes that can be reliably detected by a camera and used for pose estimation. Early work focused on designing marker geometries and classical vision pipelines that remain detectable despite turbidity, color shift and variable lighting. More recent research has extended these ideas using fiducial systems such as ArUco or AprilTag, often combined with learning-based detectors, while still relying on the geometric structure of a printed pattern for precise pose estimation.

Marie’s Stagbug AUV docking system uses a binary encoded pole as a passive black and white target mounted inside a suspended docking station. The pattern is designed so that it is visible from any arbitrary yaw, roll or pitch angle and also serves as a structural station that the AUV can grip on. Detection is performed with a fast Haar feature template matcher over the image, where the ‘dash-dot-dash’ black and white pattern is scored over multiple scales, position and image rotations to locate the camera pose relative to the dock [?]. As the AUV approaches closer, the system switches to shorter templates to maintain more precise tracking at close distance. This method purely using a high contrast pattern can deliver robust homing without any powered beacons, however, it is limited by the CPU cost on the template design as it searches over multiple scales and rotations, and restricts design flexibility for the dock station.

An alternate method uses common artificial fiducial markers like AprilTag and ArUco for pose estimation. Notably, Xu’s AUV navigation system incorporates algorithms to detect a grid of multiple markers for improved accuracy [?]. The ArUco library first decodes the marker ID and recovers each marker’s pose relative to the camera. Then, Xu’s custom noise model computes an optimal combined pose estimate from all visible markers, improving accuracy and robustness against individual detection errors. This work is framed for AUV navigation, but the principles apply directly to docking scenarios where multiple markers can be placed on the station for enhanced detection, as shown in Figure 6.

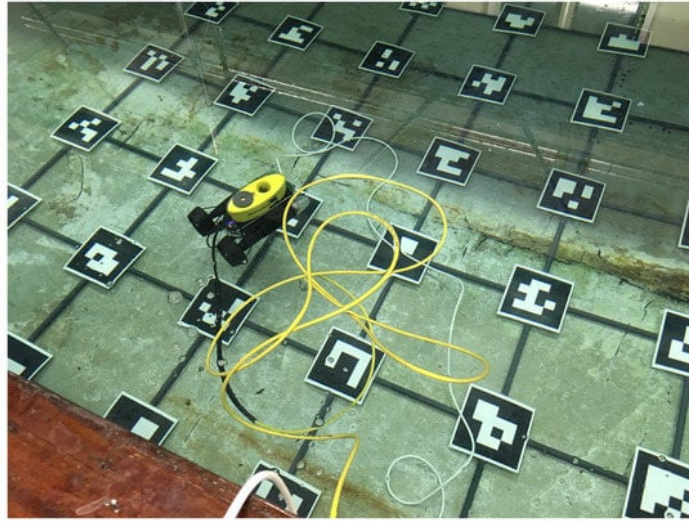


Figure 5: Xu’s AUV navigation test in a towing tank for camera pose estimation[?]

Most recently, Ni et al. proposed a hybrid active and passive docking marker where a printed AprilTag is placed at the center of the dock station, surrounded by a ring of LED lights [?], inspired by Trslic’s four light beacon active marker design [?]. A custom trained YOLO-D model is trained to simultaneously detect the LED array and the AprilTag in raw underwater images (see Figure 7).

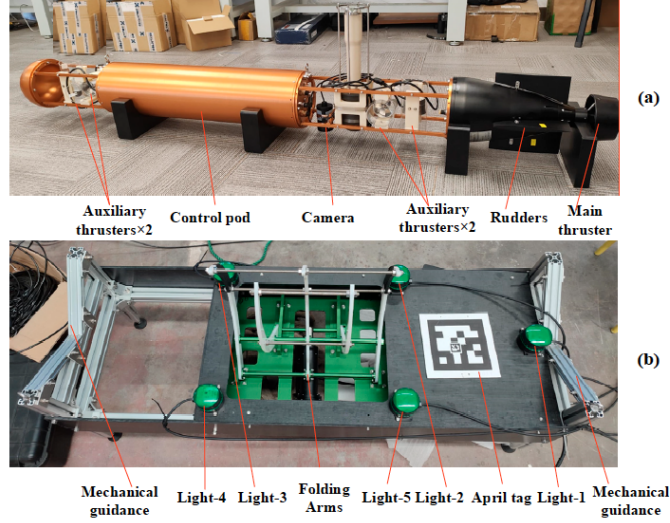


Figure 6: Example of an underwater docking station and vehicle: (a) AUV; (b) Docking station with five green LED beacons (Light-1 to Light-5), AprilTag marker at the center and v-shaped mechanical guides.[?]

Ni et al.’s AUV docking guidance is composed of 2 level of detection:

- Level I: the LED array is used as visual markers for pose estimation. However, at close range, the light scattering can introduce errors in estimating the center of the guidance light source.
- Level II: the AprilTag is used for precise pose estimation. However, this guidance method is limited by the detection range and visibility of the tag under turbidity, water quality and illumination.

The docking process begins by capturing an image, which is preprocessed through configured filters to minimise noise in the image. The YOLO-D model then detects the center X_L of the LED array and the AprilTag region of interest (ROI) in the image frame. In Level I guidance, the obtained X_L is used to compute a coarse position P_L of the docking station relative to the AUV. In Level II guidance, the AprilTag corners X_A are extracted from the ROI and used to compute a precise pose P_A via PnP. Finally, the pose estimates from both levels are fused to compute overall P_{out} , following the formula:

$$P_{out} = w_L P_L + w_A P_A$$

where w_L and w_A are weights assigned to the Level I and Level II pose estimates based on their preconfigured confidence levels. This fusion allows the system to leverage the long-range detection of

the LED array while maintaining the precision of the AprilTag when visible. The full semantic of the system is illustrated in Figure 7.

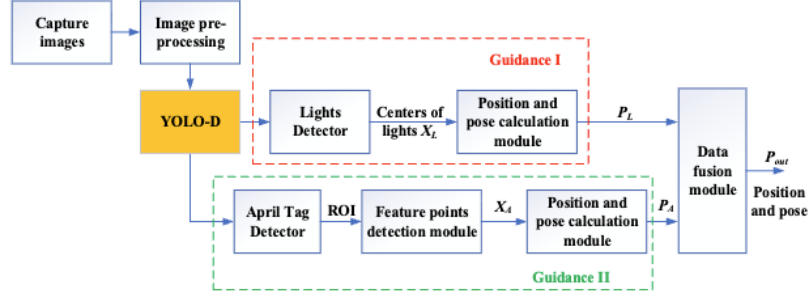


Figure 7: Ni et al.'s AUV software structure for marker to pose estimation [?].

These existing works show a clear evolution in passive marker-based underwater docking techniques, from simple high-contrast patterns to sophisticated multi-marker systems combining fiducials and active lights. Early systems like the encoded pole rely on hand-crafted patterns and feature template matching. Though their experiments demonstrated robust detection in controlled tank conditions, the computational cost and sensitivity to lighting changes limit their practical range. ArUco/AprilTag based systems are able to leverage well-established libraries for detection and pose estimation, but their performance underwater can degrade due to turbidity and lighting variability. Ni et al.'s hybrid approach combining active LED markers with AprilTags through YOLO-D detection achieves higher robustness in underwater disturbances, yet trade-offs in model size, training data and onboard compute remain challenges for real-time deployment. Overall, passive marker-based docking remains a viable and effective strategy, especially when combined with active markers and learning-based detection to overcome underwater visibility challenges.

2.2.2 Learning-Based Approaches using CNN Detection

Modern deep learning has been a game changer for underwater vision. Instead of manually designed filters or thresholding, Convolutional Neural Networks (CNNs) can be trained to recognise the docking station or its markers in complex underwater images even under low light conditions.

A prominent trend is adapting state-of-the-art detectors like YOLO (You Only Look Once) for underwater use. These one-stage detectors are fast and can work in real time on AUV hardware. However, standard YOLO models trained on everyday images perform poorly underwater due to low-light conditions [?]. Researchers have created underwater-specific variants: for example, Pan et al.(2024) proposed YOLOv9s-UI, adding custom attention modules to better extract features in underwater scenes [?]. Others integrated image enhancement into the network: Wang et al.(2023) introduced a self-localisation method using "Pseudo-3D vision-inertia for AUVs" [?] by combining a single downward camera, IMU and a depth sensor to estimate the AUV's 3D trajectory over the seabed. This is achieved by fusing visual feature tracking of the seabed with inertial and depth measurements. However, this method assumes the seafloor is locally flat and the vehicle is always at certain relative distance to the seafloor. Ni et al. (2025) introduced a specialised network called YOLO-D aimed at

docking markers [?]. YOLO-D retained the fast YOLOv11 architecture but added components for small object detection at long range, such as a BiFPN (bidirectional feature pyramid) architecture for multi-scale feature fusion and an adaptive kernel convolution (AKConv) module to improve detection accuracy and efficiency on lights and AprilTag features (see Figure 8). These enhancements significantly improved the recall of tiny distant lights or partially visible tags in testing, compared to a baseline YOLO model. In pool trials, the YOLO-D could detect and locate the LED array and AprilTag in real time, providing inputs to the docking controller even under varying lighting.

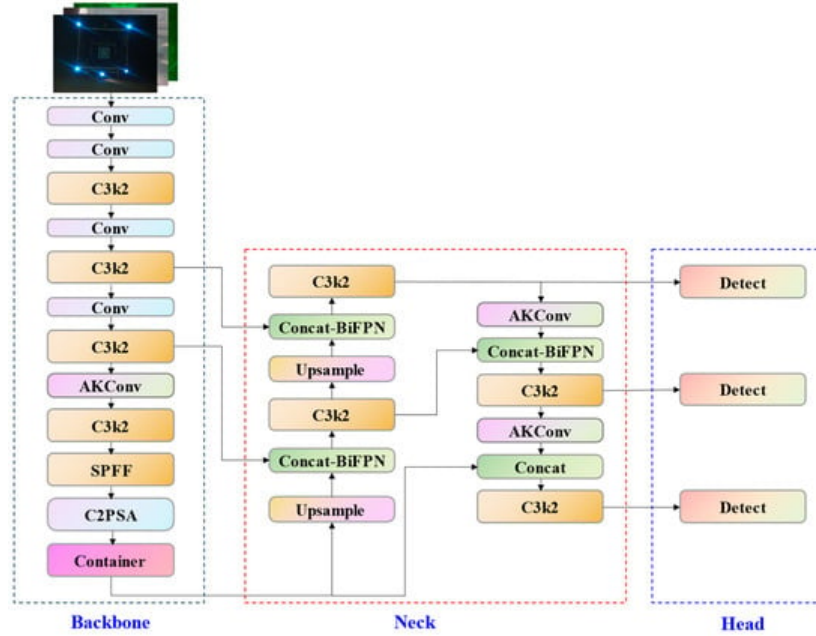


Figure 8: Ni et al.'s YOLO-D network architecture[?].

Beyond detection, some learning approaches go further by performing end-to-end control or segmentation. One innovative method used a CNN to directly output the steering command needed for docking from the camera image. Sans-Muntadas et al. (2019) tackled on the lack of underwater precise positioning systems by proposing a CNN that uses raw image from the onboard camera to estimate the heading error relative to the docking station, which is then fed to the controller to maneuver to the docking station [?]. This approach does not require explicit marker detection or pose estimation or other external sensors. However, their system produces false estimates when the dock is close to the docking station due to the lack of visible features in the close-up view.

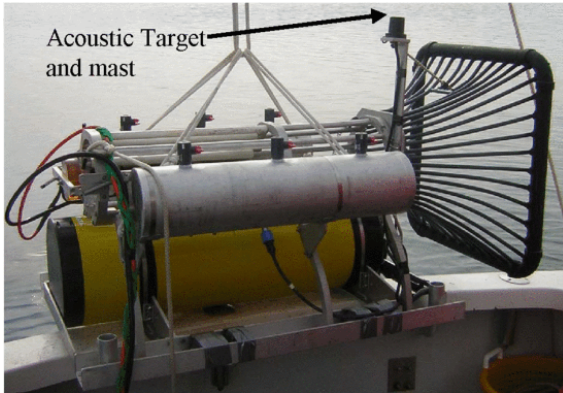
In summary, learning-based methods (especially CNN detectors like YOLO variants) offer speed and robustness for vision-based docking. They can detect docking targets in complex scenes where classical methods fail. Yet, they often serve as part of a larger system, for example, a CNN might detect an AprilTag's bounding box, but then a precise pose is computed by a PnP algorithm on its corners. Also, pure CNN approaches for control are still experimental.

2.3 Other Sensing Methods and Hybrid Systems

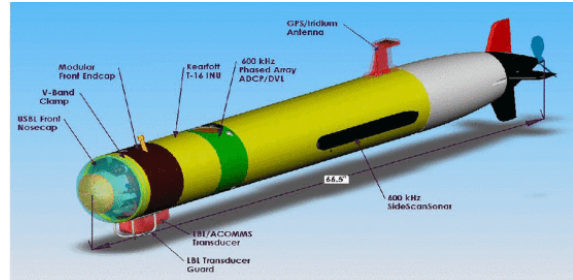
Beyond vision, other sensing methods have been explored for underwater docking systems, including acoustic navigation, magnetic / EM homing, and mechanical alignment aids.

2.3.1 Acoustic positioning and homing

Acoustic based navigation system is widely used as a long range communication underwater for docking. A common configuration has an acoustic transceiver on the dock that emits a signal, which is received by a transponder on the AUV. The AUV then replies with its own signal, allowing the dock to compute the distance based on the time delay. To obtain bearing information, several inverted ultra-short baseline (iUSBL) receivers are spaced precisely on the AUV hull. By measuring the slight time difference of arrival of the acoustic signals at each receiver, the system can triangulate the relative azimuth and elevation angles between DS and AUV. WHOI's REMUS AUV docking system is a prime example of a USBL-navigation aided system[?] (see Figure 9a and 9b). This acoustic homing can guide the AUV from long range (up to 2km), at the cost of moderate update rates, latency due to speed of sound underwater, risk of false signals from acoustic refraction/ambient noises and limited angular resolution.



(a) Remus-100 AUV docking station with labelled acoustic transceivers [?].



(b) Remus-100 AUV docking vehicle with labelled components. Note the USBL receivers at the nose-cap [?].

Figure 9: Remus-100 docking station and vehicle (left: station; right: vehicle) [?].

In practice, many working-class AUVs (REMUS, Bluefin, etc.) combine iUSBL range/bearing with Doppler Velocity Logger (DVL) and inertial navigation to accomodate for current-induced drift and improve overall localisation accuracy during the approach phase.

2.3.2 Magnetic / Electromagnetic Homing

Electromagnetic (EM) guidance offers a more robust alternative to acoustic signal with low-frequency magnetic or electro-magnetic fields emitted by a coil on the dock, resulting in lower latency. Feezor et al. (2001) developed an EM homing system where a low-frequency magnetic field is generated by a coil on the dock, and the AUV uses magnetometers to detect the field strength and direction, without line-of-sight constraints [?]. Their experiments demonstrated that the AUV could home in on the dock

from a limited range up to 30m. Further, the system requires dedicated coils, drivers and calibration of the local field environment, which may limit its practicality for general-purpose docking stations.

2.3.3 Mechanical Alignment Aids

Most underwater docking stations incorporate mechanical alignment aids to relax the sensing requirement and accommodate for the slight tolerances during the final docking alignment phase. There exist two main configurations: unidirectional and omnidirectional guides.

Unidirectional docking stations are the most common type of configuration, typically designed for torpedo-shaped AUVs that approach from a known heading. These stations feature funnel/cone shaped guides to help channel the vehicle into the dock despite some slight misalignment. However, this design imposes condition on the AUV to know at all time the pose of the dock relative to itself, which may not be feasible for free-floating docks as the pose and location of the dock are variables.

Omnidirectional docking stations, on the other hand, are designed to allow approach from any heading. These stations often feature a vertical structure composed of poles or cables under tension, allowing the vehicle to be latched onto the dock regardless of its orientation, notable in Odyssey IIb AUV's dock station design (see Figure 10a and 10b). This design simplifies the navigation system and alignment process as the vehicle only requires knowledge of the dock's location. However, omnidirectional designs require complex mechanical design on the vehicle's nose and may not fit the requirement for all AUVs as the nose is often reserved for sensors or other equipment, and may also introduce drag on the vehicle during underwater operations.

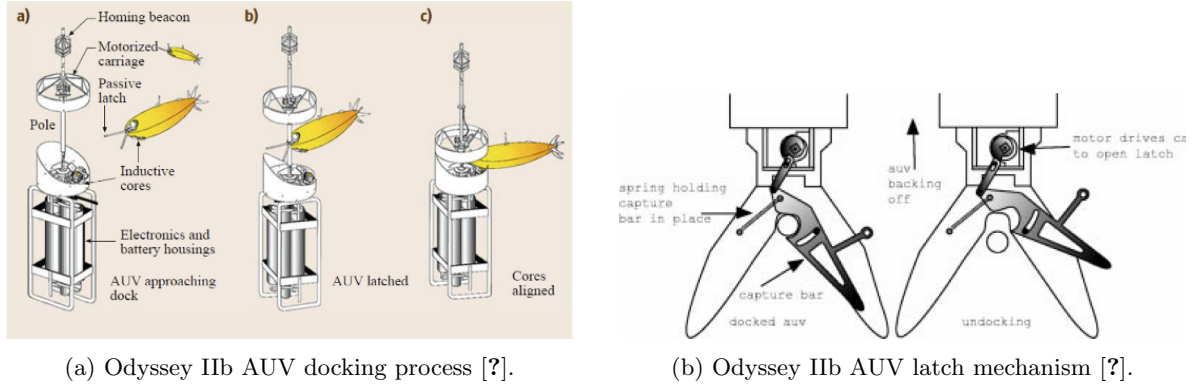


Figure 10: Odyssey IIb docking process and latch mechanism (left: process; right: latch mechanism) [?].

2.4 Addressing Known Limitations in Academic Research

Despite substantial progress in AUV docking research, several recurring limitations appear across the literature. First, most systems assume that the docking station is either seabed-mounted or fixed in a known position and orientation. Examples include seabed poles with acoustic and visual aids [?], fixed funnel-type garages for REMUS-class vehicles [?], and rigid TMS cages for work-class ROVs [?]. In these scenarios, the dock is effectively treated as static in the world frame, and the control problem reduces to steering the vehicle into a stationary target. Free-floating platforms with their own

wave-induced motion are rarely modelled explicitly, leaving a gap for docking scenarios where both the vehicle and the dock are moving bodies.

A second limitation is the experimental environment. Many vision-based docking studies are evaluated in clear-water laboratory tanks, where turbidity, backscatter, and illumination can be managed or only coarsely varied. Liu et al.’s DoNN-based cone docking, Ni et al.’s YOLO-D experiments, and various ArUco/AprilTag trials all report promising performance in tanks or test pools, but only limited results in truly turbid or highly dynamic field conditions. Quantitative evaluation across different turbidity levels, illumination levels, and camera-dock distances is often missing, making it difficult to assess how methods will perform in more challenging operational settings.

Finally, while deep learning has improved robustness in visual detection, many proposed architectures assume access to relatively powerful onboard compute and extensive labelled datasets. Underwater-specific YOLO variants, segmentation-assisted pipelines, and pseudo-3D visual-inertial systems often rely on GPU-class hardware and offline training using curated datasets. Real-time adaptation to changing turbidity or lighting is usually handled via static data augmentation or pre-configured enhancement modules rather than closed-loop adaptation. Likewise, few studies explicitly address the constraints of small or mid-size ROV hardware, where power, compute and communication bandwidth are more limited.

2.5 How This Project Fits Into the Literature

This project is positioned to address several of the above gaps by focusing on a vision-based docking system for a low cost ROV approaching a free-floating underwater dock. Unlike many prior works that assume a rigid, seabed-mounted or fixed-frame dock, the proposed system will explicitly consider a docking station that is suspended in the water column and mechanically coupled to a rigid structure. This introduces relative motion between the dock and the ROV, even when both attempt to hold position, and more closely reflects resident ROV scenarios and practical offshore deployments.

On the perception side, the project builds directly on fiducial-based and learning-based detection approaches identified in the literature review. The dock will be equipped with passive markers (e.g., AprilTags or ArUco patterns) and simple active LEDs, enabling comparison between classical marker detection and lightweight CNN-based detectors inspired by recent Ni et al.’s underwater docking work. Rather than attempting to reproduce full-scale architectures such as YOLO-D, the emphasis will be on achieving robust detection and pose estimation within the compute envelope of an ROV platform, and on characterising how performance degrades under changes in turbidity, lighting and relative orientation.

Experimentally, the project aims to bridge the gap between idealised tank tests and more realistic field-like conditions. Controlled experiments will be conducted in a test environment where water clarity, lighting and dock motion can be varied in a repeatable way. Key evaluation metrics will include marker detection rate, pose estimation error, docking success rate, and latency from image capture to control command. By systematically varying environmental and hardware parameters, the project will generate practical guidelines on marker design, placement, and algorithm selection for ROV docking.

In summary, this thesis contributes to the docking literature by formulating and tackling the prob-

lem of vision-based docking to a free-floating underwater dock, tailoring and evaluating fiducial and learning-based perception methods for ROV-class hardware and constraints, and providing experimental evidence and design recommendations that help translate academic docking techniques into more realistic ROV deployment scenarios.

3 Research Question and Project Plan

3.1 Primary Research Question

What is the feasibility and performance of a vision-based, turbidity-adaptive pose estimation system for autonomous docking of an ROV to a free-floating underwater station?

3.2 Sub-Questions

The primary question can be decomposed into several sub-questions:

1. How robust are fiducial-based (AprilTag/ArUco) and learning-based (YOLO) detection methods under varying turbidity levels, lighting conditions and relative orientations between ROV and dock?
2. What pose estimation accuracy and update rate are required for safe docking, given the relative motion of a free-floating dock and the dynamic response of the ROV and tether?
3. How can motion compensation and control laws be designed to account for dock motion while still precisely maneuvering to the docking pose?

3.3 Hypothesis and Expected Feasibility

The working hypothesis for this project is:

A turbidity-adaptive, two-level vision pipeline (light cues for long-range guidance and markers for close-range precision), combined with appropriate motion compensation and simple fallback beacons, can achieve reliable short-range docking of an ROV to a free-floating underwater station in conditions of moderate turbidity.

In particular, it is expected that a combination of image enhancement, robust fiducial detection, light-based guidance will provide accurate pose estimates to guide the ROV through the final approach and alignment phase. However, high turbidity levels that completely blocked camera’s visibility are beyond the capability of vision alone. In such cases, hybrid sensing (e.g. acoustic or EM navigation system) or manual operator intervention will likely be required.

3.4 Proposed Methodology Overview

The proposed methodology consists of three main elements: perception, pose estimation and control.

Perception and Dataset Collection The perception pipeline will be developed from the outset using real images from the ROV camera system. Initially, a benchtop camera rig and small tank setup will be used to capture images of the docking station mock-up equipped with LEDs and ArUco/AprilTag markers under varying distances, orientations and lighting. As hardware becomes available, the BlueROV camera will be used to collect sequences in a pool, including both static and gently moving dock configurations.

An image pre-processing stage will be implemented to perform basic filtering and thresholding to enhance marker and light visibility. This module will be tuned empirically using these real datasets under different turbidity levels to balance noise suppression against improved marker and light visibility.

Two-Level Vision Guidance and Pose Estimation On top of pre-processing, a two-level detection pipeline will be implemented:

- **Level I (light-cue guidance):** detection of LED arrays for long-range, coarse guidance, using a lightweight CNN (YOLO) for robust light-cluster localisation.
- **Level II (marker guidance):** detection of passive AprilTag/ArUco markers mounted on the dock for close-range, high-precision guidance, using established python libraries and a CNN-based ROI detector as backup under degraded conditions.

Pose estimation will be performed via Perspective- n -Point (PnP) solvers with RANSAC outlier rejection, using known 3D geometry of the light cluster and marker and their 2D image measurements. Separate light-based and marker-based pose estimates will be fused in a temporal filtering framework (e.g. a Kalman or complementary filter) to provide a smooth, confidence-weighted relative pose. All accuracy and robustness metrics (e.g. position/orientation error, detection range, latency) will be quantified using real imagery with ground-truth or pseudo ground-truth measurements (e.g. calibrated test rigs and known distances).

Control and Motion Compensation (Simulation-Focused). Control and motion-compensation strategies will be developed primarily in Gazebo/ROS 2, using an abstract “pose sensor” that provides relative dock pose to the simulated ROV. The noise characteristics, latency and dropout behaviour of this sensor will be configured based on statistics measured from the real-image perception pipeline.

In sim, the free-floating dock will be modelled with simple sway or heave motion to emulate cable-suspended behaviour. Using the abstract pose feed, a closed-loop controller will be designed to manage:

- *Coarse approach* along the dock axis using Level I pose primarily;
- *Fine alignment* and position hold using Level II pose primarily;
- *Final engagement* into the docking structure within allowable tolerances;
- *Abort and fallback behaviours* when pose confidence or update rate drops below thresholds.

Controller gains, mode-switching logic and safety conditions will be tuned entirely in Gazebo until repeatable simulated docking is achieved over a range of initial conditions and dock motions. Because the perception stack is not simulated, this stage isolates the control problem from rendering artifacts and keeps simulation focused on dynamics.

Integration with the Physical BlueROV Once both components are mature, the perception and pose-estimation stack will be deployed on the BlueROV platform and connected to the control loop. Initial tests will be conducted with conservative behaviours (slow approach, generous abort margins) in a pool with a static dock, before progressing to experiments with a gently moving, suspended dock. Experimental results will then be used to refine both the pose error model (fed back into simulation) and the controller tuning.

3.5 System Block Diagram Description

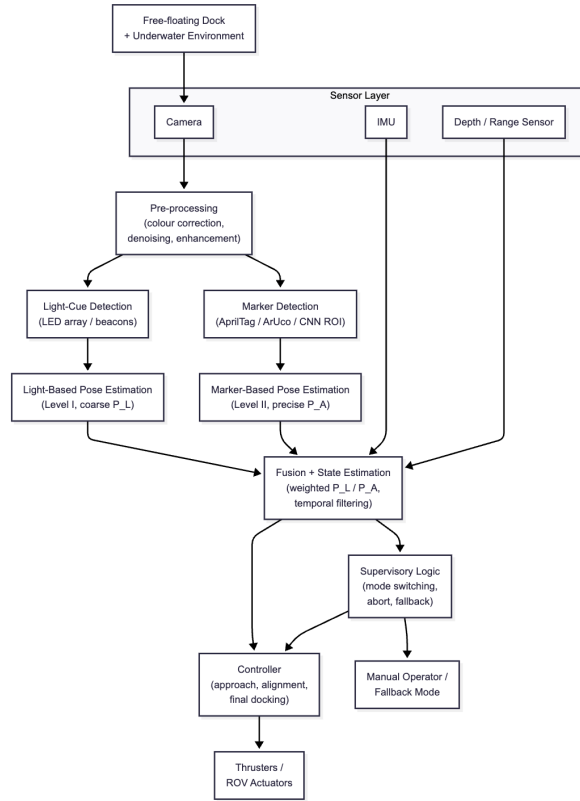


Figure 11: High-level system block diagram showing the two-level visual guidance pipeline (light cues for long-range coarse guidance and markers for close-range precision), pose fusion and state estimation, and closed-loop control for ROV docking to a free-floating underwater station.

3.6 Timeline

The core technical development of this project will be undertaken during Thesis B and Thesis C. Thesis A is limited to preliminary work such as basic ArUco-based pose estimation on real images, basic CAD modelling of the docking station and detailed planning.

Thesis B

- **Weeks 1–2: Perception pipeline consolidation.** Finalise the basic image acquisition and pre-processing pipeline using the ROV camera and benchtop/tank setups. Implement and test the two-level detection scheme (Level I light cues, Level II markers) on real underwater images, and verify that PnP-based pose estimation runs reliably offline.
- **Weeks 3–4: Quantitative evaluation on real imagery.** Collect structured datasets in a pool or tank with a static dock under varying distances, orientations, turbidity and lighting. Quantify detection range, pose accuracy (position/orientation error) and latency for both light-based and marker-based guidance. Derive empirical statistics for pose noise, dropout and update rates.
- **Weeks 4–6: Control development in simulation (Gazebo/ROS2).** Configure a simplified Gazebo scenario with a BlueROV-style model and the designed free-floating dock. Design and tune the docking controller (coarse approach, fine alignment, abort/fallback logic).
- **Weeks 6–8: Initial integration and short pool trials.** Deploy the perception and pose-estimation stack on the ROV compute platform and connect its output to a conservative version of the controller. Perform short pool trials with a static dock to verify that real-time pose feeds satisfy the assumptions used in simulation (e.g. update rate and noise bounds) and that basic approach/alignment behaviours are stable.
- **Week 9: Reporting and planning for Thesis C.** Finalise the Thesis B report, summarising: (i) perception performance on real underwater imagery; (ii) controller performance in Gazebo under realistic pose-feed emulation; and (iii) identified gaps and risks for full integrated docking in Thesis C.

Thesis C

- **Weeks 1–3: Full-system integration and static-dock docking.** Refine the integration between perception, state estimation and control on the physical ROV. Conduct systematic pool trials with a static dock to demonstrate repeatable short-range docking, measure success rate and time-to-dock, and adjust safety margins and abort thresholds.
- **Weeks 3–6: Free-floating dock experiments.** Introduce a suspended or otherwise movable dock to emulate free-floating behaviour (e.g. cable-suspended frame with induced sway). Evaluate how the system handles relative motion and tether effects, and refine motion-compensation strategies if required. Update the pose-error and delay model and, if necessary, retune the controller in Gazebo.
- **Weeks 6–8: Robustness and envelope characterisation.** Perform extended test campaigns with multiple initial conditions, repeated runs, and varied turbidity/lighting levels. Quantify docking success rate, time-to-dock, and any failure modes. Identify the operational envelope (e.g. maximum turbidity, allowable dock motion) within which vision-based docking remains reliable.

- **Weeks 8–10: Final analysis, recommendations and reporting.** Consolidate all experimental results, compare simulated and real-world behaviour, and formulate design recommendations for vision-based ROV docking to free-floating stations. Complete the Thesis C report and supporting artefacts (figures, videos, code documentation) for submission and potential future extension.

The exact week boundaries may be adjusted according to equipment availability and academic calendar constraints.

4 Project Dependent Preparations

This section summarises the project-specific preparations completed or in progress at the time of this report. The focus for Thesis A is on building a minimal but functional vision pipeline using ArUco fiducials on real images, and developing an initial CAD model of the free-floating docking station. These activities provide early evidence that the proposed methodology is practical and that the broader Thesis B/C timeline is achievable.

4.1 Setting up Software Environment

To support the planned vision-based docking system, the following software tools and libraries have been set up:

- **Python and OpenCV:** A Python 3 environment has been configured with the OpenCV and `opencv-contrib` libraries for image capture, pre-processing and ArUco marker detection. Basic camera access and image display have been verified on the target machine.
- **ArUco pose estimation tools:** The OpenCV ArUco module has been installed and a small calibration and pose-estimation pipeline has been developed. This includes camera calibration using a chessboard pattern, marker detection, and Perspective- n -Point (PnP) pose estimation with visual overlays on the image.
- **ROS 2 Humble workspace:** A ROS 2 Humble workspace has been created to host the future ROV control stack. Within this workspace, a prototype “vision node” package has been started, intended to wrap the ArUco pose-estimation pipeline and publish the estimated dock pose as a ROS 2 message for later use by the controller.
- **CAD tooling:** A parametric CAD package (e.g. SolidWorks / Fusion 360) has been selected for modelling the docking station, and an initial project structure has been created for the dock geometry, mounting points and marker/light locations.
- **Version control:** A private Git repository has been created to store Python scripts, ROS 2 packages, CAD files and experimental configuration. This enables reproducibility and incremental development across Thesis B and C.

4.2 Preliminary ArUco Pose Estimation Prototype

As an initial proof-of-concept, a basic ArUco-based pose estimation pipeline has been implemented using real camera images. The goal at this stage is not full docking, but to verify that:

1. ArUco markers can be reliably detected under underwater-like lighting conditions, and
2. the resulting Perspective- n -Point (PnP) pose estimates behave sensibly as the camera moves.

The prototype currently consists of two main components. First, a standalone Python calibration script is used to estimate the intrinsic parameters and distortion coefficients of the camera from a set of checkerboard images. This script runs offline (outside ROS 2) and outputs a calibration file (`camera_calib.npz`) containing the camera matrix and distortion coefficients that are later loaded by the vision node at runtime.

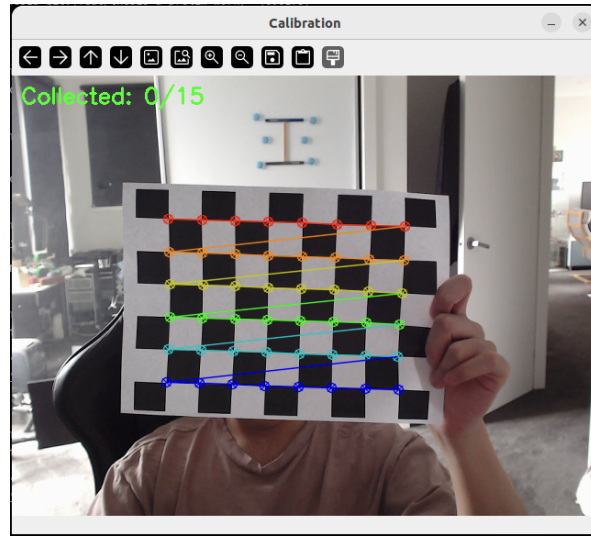


Figure 12: Example of sampling frames used for camera calibration with a chessboard pattern.

Second, the core ArUco-based detection and pose estimation pipeline has been implemented as a ROS 2 Humble node. This node subscribes to a camera image topic (e.g. `/camera/image_raw`), applies the pre-computed calibration, detects ArUco markers in each frame, and computes their 6-DoF pose using OpenCV's PnP solver. Detection overlays (marker borders and coordinate axes) are rendered into a debug image stream, and the estimated translation and orientation of the marker are published as a `PoseStamped` message on a dedicated topic (e.g. `/dock_pose`). A simplified view of the current node setup for this prototype is shown in Figure 13.

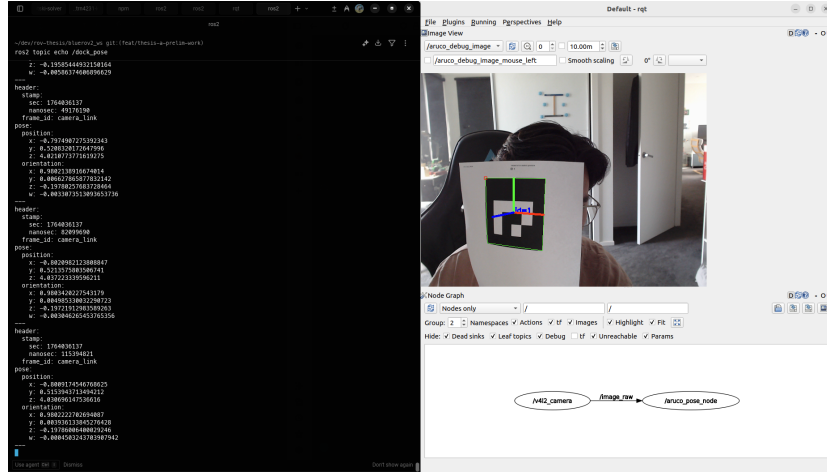


Figure 13: Initial ROS 2 Humble vision node prototype: the ArUco-based pose-estimation node subscribes to camera images, applies calibration and publishes an estimated dock pose topic for use by control and logging modules.

These preliminary results demonstrate that marker-based pose estimation can be implemented and integrated into the ROS 2 environment on the target hardware. In subsequent thesis stages, this prototype will be extended to handle multiple markers, underwater imagery, turbidity-aware pre-processing and tighter coupling with the ROV's navigation and control stack.

4.3 Initial Docking Station CAD Model

An initial concept CAD model of the free-floating docking station has been developed, focusing on:

- **Marker placement:** Locations for ArUco fiducials arranged to maximise visibility from multiple approach angles, considering occlusions and perspective distortion.
- **Active lighting:** Integration points for LED beacons to provide long-range guidance in low-visibility conditions.

The dock station is designed to be suspended from a floating platform (see Figures 14a and 14b).

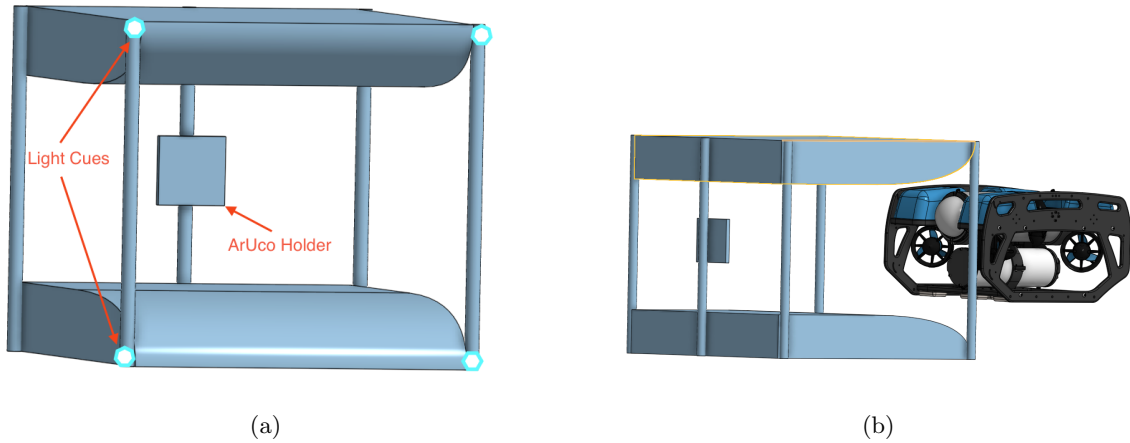


Figure 14: Concept CAD views: (a) docking station with marker and LED placements; (b) docking station with BlueROV2 for scale.

Conclusion

This report has defined and scoped a vision-based docking system for an inspection-class ROV approaching a free-floating underwater station, motivated by the need to reduce reliance on risky, vessel-dependent launch and recovery.

Through a literature review, it has surveyed acoustic, electromagnetic, mechanical and vision-based docking methods, and identified key gaps: most work assumes seabed-mounted or otherwise static docks, focuses on AUVs rather than tethered ROVs, and is validated mainly in clear, controlled conditions rather than turbid, dynamic environments. In response, the project has been framed around a turbidity-adaptive, marker-based pose estimation pipeline for a BlueROV-style platform, with a proposed architecture, aiming to enhance the feasibility for low cost autonomous underwater operations.

Within Thesis A, foundational preparations have been completed: the software environment and ROS 2 Humble stack have been configured, camera calibration performed, and a working ArUco-based pose estimation node and initial dock CAD concepts implemented. Next Thesis B and C work will extend this prototype to underwater imagery and turbidity-aware pre-processing, integrate it with the ROV's navigation and control, and experimentally evaluate autonomous docking performance under representative conditions.

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